외계행성 대기
Exoplanetary Atmospheres

Lecture 8: Vertical Thermal Structure of a Planetary Atmosphere II
2026 November 2 (Monday)

Based on: Chapter 9 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 4 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 8

Surface Temperature in Simple Greenhouse Models (Continued)

Convection versus Radiation

The Radiative-Equilibrium Temperature Profile

Greenhouse and Anti-Greenhouse Effects

The Adiabatic Temperature Profile

The One-Dimensional T–P Profile

Temperature Retrieval

Summary

A leaky greenhouse: the setup

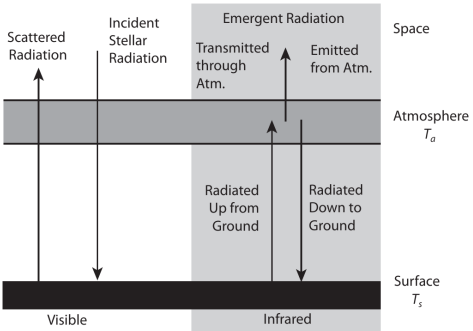


Figure 1: The leaky greenhouse: the layer absorbs only a fraction α of the infrared. Seager (2010), Fig. 9.5.

The simple greenhouse atmosphere traps *all* infrared radiation from the ground. A slightly more realistic "leaky" layer is partly transparent (optically thin) in the infrared:

- the atmosphere layer absorbs a fraction $\alpha \in (0,1)$ of the upwelling surface emission and transmits $1-\alpha$ to space;
- by **Kirchhoff's law** its emissivity equals its absorptivity$\times$B, so it emits the flux $\mathscr{F}_{\text{emiss}}\uparrow = \alpha\sigma T_a^4$ upward and $\mathscr{F}_{\text{emiss}}\downarrow = \alpha\sigma T_a^4$ downward;
- starlight is again assumed to pass the layer and heat the ground. Again, the starlight is assumed not to be scattered back to space and all of the radiation reaches the planet surface.

A leaky greenhouse: the solution

Two contributions now leave the planet (Fig. 9.5):

- emission from the atmosphere layer;
- surface emission that **leaks** through the layer.

Two energy-balance equations (top and surface) determine the two unknowns T_s and T_a:

$$\begin{aligned} \text{top of the atmosphere:} \quad & \mathscr{F}_* \downarrow \quad = \mathscr{F}_{\text{atm}} \uparrow + (1-\alpha)\,\sigma T_s^4 \uparrow, \\ \text{surface:} \quad & \mathscr{F}_* \downarrow + \mathscr{F}_{\text{atm}} \downarrow = \sigma T_s^4 \uparrow . \end{aligned}$$

Here, $\mathscr{F}_* \downarrow = \sigma T_e^4$ is the stellar flux absorbed by the ground and $\mathscr{F}_{\text{atm}} \uparrow = \mathscr{F}_{\text{atm}} \downarrow = \sigma T_a^4$ is the flux emitted by the atmosphere layer.
Eliminating $\mathscr{F}_{\text{atm}} \downarrow = \mathscr{F}_{\text{atm}} \uparrow$: $2T_e^4 = (2-\alpha)\,T_s^4$. Then $T_a^4 = T_s^4 - T_e^4 = \alpha T_e^4/(2-\alpha)$:

$$\boxed{T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e}, \qquad \boxed{T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e = \left(\frac{\alpha}{2}\right)^{1/4} T_s}.$$

Check: the atmosphere layer absorbs $\alpha\sigma T_s^4$ and emits $2\sigma T_a^4$, so $T_s^4 = (2/\alpha) T_a^4$.

A leaky greenhouse: what it means

$$T_s = \left(\frac{2}{2-\alpha}\right)^{1/4} T_e, \qquad T_a = \left(\frac{\alpha}{2-\alpha}\right)^{1/4} T_e$$

- A partly transparent layer **reduces** the greenhouse warming.
- $\alpha = 1$: the simple greenhouse, $T_s = 2^{1/4} T_e$ (a special case of the leaky model). $\alpha \to 0$: transparent layer or no atmosphere, $T_s = T_e$ and $T_a = 0$ (no atmosphere).
- Since $1 \leq 2 - \alpha \leq 2$: always $T_s \geq T_e \geq T_a$. The ground is warmer, and the atmospheric layer colder, than the emission temperature is the lowest.
- Earth: $T_s = 288\,\mathrm{K}$ with $T_e = 255\,\mathrm{K}$ needs $2/(2-\alpha) = (288/255)^4 \approx 1.63$, i.e. $\alpha \approx 0.77$.
- **A single layer can never give more than $T_s = 2^{1/4} T_e$. Venus needs many opaque layers** (next slide). Extended to many layers, with the right α for each layer at every wavelength, and with molecules and clouds, the model approaches a multilayer **radiative-equilibrium** model.

Many ‘non-leaky’ layers: towards radiative equilibrium

Stack n layers above the surface (Seager’s exercise 3). Assumptions:

- All layers are **identical in their optical properties**: transparent in the visible (starlight heats only the ground) and **fully opaque in the infrared** ($\alpha = 1$), each emitting as a black body up and down.
- Only radiation carries energy (no scattering, convection or conduction). Thickness, density and pressure never enter; the layers differ *only* in temperature, which is an output of energy balance, not an input.

Number the layers $k = 1, \ldots, n$ from the top, with the surface as $k = n+1$ and $T_0 = 0$ (no infrared from space). Layer k absorbs the emission of its two neighbors and emits from both faces, $2\sigma T_k^4 = \sigma T_{k-1}^4 + \sigma T_{k+1}^4$, with $\sigma T_1^4 = \sigma T_e^4$ (the top layer is what space sees). T_k^4 is therefore linear in k:

$$T_k^4 = k\,T_e^4 \quad \Longrightarrow \quad \boxed{T_s = (n+1)^{1/4}\,T_e} \quad \text{(for } n \text{ opaque layers with } \tau_k \approx 1 \text{ and } \tau_{\text{tot}} \approx n\text{)}.$$

- Venus: $(T_s/T_e)^4 = (730/230)^4 \approx 100$, i.e. the equivalent of ~ 100 opaque layers — an infrared optical depth of order 100 above the surface. The massive CO_2 atmosphere (92 bar) is that opaque.
- With $n \to$ many layers of appropriate absorptivity, at all wavelengths, with molecules and clouds, these models become a **multilayer radiative-equilibrium model**: the subject of Lecture 8.
- Each “layer” is one unit of infrared optical depth, not a fixed geometric thickness. The model is a discrete gray atmosphere with $\tau \approx n$, and $T_k^4 \propto k$ mirrors the gray $T^4 \propto \tau$.
- For leaky layers ($\alpha < 1$) the recursion changes and $T_s < (n+1)^{1/4} T_e$; already for $n = 1$, $T_s^4 = 2T_e^4/(2-\alpha)$.
- The one-layer ratio $T_a/T_s = 2^{-1/4}$ reappears in the gray model as the skin temperature $T(0) = 2^{-1/4} T_{\text{eff}}$.

Convection versus Radiation: Which process carries the energy?

Energy can be transported by **radiation, convection, or conduction**. In the relatively dense lower atmospheres, thermal conduction is negligible. Energy takes the **path of least resistance**: if the opacity is low, then the energy is most likely to be transported by photons; if the opacity is high, radiation is impeded and energy is more likely to be transported by convection.

How the temperature gradient determines whether or not convection occurs:
Stability of a displaced parcel. Lift a parcel slightly: it expands *adiabatically* to the ambient pressure, keeping its own temperature and density. (adiabatic means no heat is exchanged with the surroundings.)

- if it is colder (denser) than its new surroundings: it sinks back. The atmosphere is **stable** against convection.
- if it is warmer (lighter) than its surroundings: buoyancy keeps it rising. The atmosphere is **unstable**; convection occurs.

A situation is **stable** if, after a disturbance, the system returns to its original state. Which case occurs is decided by comparing two temperature gradients (next slide).

The criterion for convection

Two temperature gradients compete:

- the **adiabatic** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{ad}}$, followed by the rising parcel;
- the **atmospheric** gradient $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ of the surroundings.

Start the parcel at height z with the ambient temperature T_0 and lift it by Δz. Both gradients are negative where T falls with height, so

$$T_{\mathrm{parcel}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}} \Delta z, \qquad T_{\mathrm{atm}} = T_0 - \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} \Delta z.$$

The parcel has the ambient pressure, so by the ideal gas law ($\rho = \mu_m m_{\mathrm{H}} P/kT$, same μ_m) **the warmer gas is lighter**. The parcel keeps rising if $T_{\mathrm{parcel}} > T_{\mathrm{atm}}$, so **the criterion for convection is a criterion for buoyancy**:

$$\boxed{\left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{atm}} > \left|\frac{\mathrm{d}T}{\mathrm{d}z}\right|_{\mathrm{ad}}}$$

(Schwarzschild criterion; Seager Eq. 9.23).

- Unstable: the atmosphere cools with height *faster* than the adiabat; the adiabat is the "shallower" gradient.
- Stable: the atmosphere cools more slowly than the adiabat, is isothermal, or warms with height (an inversion). Note that $T_{\mathrm{atm}} = T_0 + |\mathrm{d}T/\mathrm{d}z|_{\mathrm{atm}} \Delta z$ when temperature increases with height.

Stable and unstable profiles

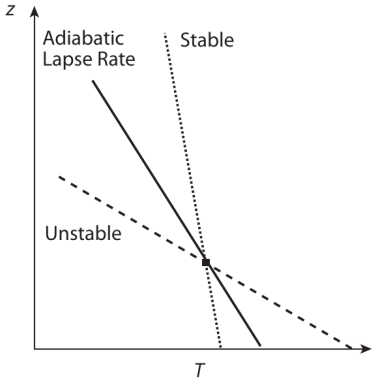


Figure 2: Profiles stable (dotted) or unstable (dashed) against convection, relative to the adiabat. Seager (2010), Fig. 9.6.

Reading Fig. 9.6 (T horizontal, z vertical), starting from a common point on the adiabat (solid line):

- a profile that loses *less* temperature per unit height than the adiabat: a lifted parcel ends up colder and denser than its surroundings and sinks back: **stable**;
- a profile that loses *more* temperature per unit height than the adiabat: the lifted parcel is warmer and lighter and keeps rising: **unstable**.

An unstable profile cannot persist: convection mixes the layer and drives its gradient back towards the adiabat. Convective regions therefore follow the adiabat closely (Lecture 8).

To decide *when* a layer is unstable we need the actual sizes of the two gradients (next slide).

When is the adiabatic gradient small, or the radiative gradient large?

Which $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}}$ goes into the criterion? We ask whether a layer in which *radiation alone* carries the flux would be stable. Its gradient is then the radiative-equilibrium one, $(\mathrm{d}T/\mathrm{d}z)_{\mathrm{atm}} \approx (\mathrm{d}T/\mathrm{d}z)_{\mathrm{rad}}$. If that gradient is unstable, convection sets in and the actual gradient relaxes towards the adiabat.

The two gradients to compare (Seager Eqs. 9.24–9.25):

$$\left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{ad}} = -\frac{g}{c_p}, \qquad \left(\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{rad}} = -\frac{3}{16}\,\frac{\kappa F_{\mathrm{int}}}{\sigma T^3} \quad \Longrightarrow \quad \text{convection if} \quad \frac{3\kappa F_{\mathrm{int}}}{16\sigma T^3} > \frac{g}{c_p}.$$

The adiabat is derived in Lecture 8. The radiative gradient is the Rosseland diffusion result of Section 1, with κ the mean extinction per unit length (α_R).

- **Small adiabatic gradient = large c_p.** c_p is the energy needed to heat unit mass of gas by 1 K at constant pressure. Where hydrogen ionizes, the added heat goes into removing electrons rather than into the kinetic energy of the gas, so c_p becomes large and g/c_p small: ionization zones of stars, and the hot interiors of giant planets like Jupiter, where hydrogen starts to ionize above ~ 4000 K.
- **Large radiative gradient = large F_{int} or large κ.** The diffusion formula holds only deep in the atmosphere, but it shows the trend. Large fluxes drive the convective interiors of giant planets; in the atmospheres the flux alone is usually not large enough, and the **opacity** is the main driver (next slide).

Seager gives the units of c_p as J kg^{-1} s^{-1}; they are J kg^{-1} K^{-1}.

Convection in planets

- **Opacity drives convection.** Opacity grows with the density of absorbers, so the deep, optically thick layers are almost always convective, while radiation dominates higher up. Opacity can also change with T and P as the composition changes.
- **Jupiter:** a radiative zone in the upper atmosphere and a convective zone in the lower atmosphere — but perhaps another radiative zone underneath, where the mean opacity has a minimum ($T \approx 1300$–$1800\,\mathrm{K}$; Guillot et al. 1994).
- **Hot Jupiters:** strong irradiation makes the dayside temperature profile nearly isothermal or even inverted, so the radiative zone extends *deep* into the atmosphere.
- **Rocky planets with thin atmospheres:** most visible-wavelength starlight is absorbed at the planet surface, which becomes much hotter than the air above. The large upward flux from the ground makes the radiative gradient steep, and a **convective layer** forms just above the surface — on Earth most of the troposphere.
- **Inversions are very stable:** a lifted parcel cools as it expands while its surroundings get warmer with height, so it sinks back strongly. Stratospheres do not convect. Most of the solar system planets have stratospheres, heated by UV absorption in ozone (Earth) or hydrocarbons (Jupiter, Saturn, Titan).

Radiative equilibrium

If energy is carried only by radiation, no energy is created or destroyed in a layer: what enters a volume must leave it. Equivalently, the net flux is constant,

$$\frac{\mathrm{d}F(\tau)}{\mathrm{d}\tau} = 0.$$

Total absorption per volume is $\int\!\!\int \kappa_\nu I_\nu \,\mathrm{d}\Omega\,\mathrm{d}\nu = 4\pi \int \kappa_\nu J_\nu \,\mathrm{d}\nu$; with isotropic emission and $\varepsilon_\nu = \kappa_\nu S_\nu$, total emission is $4\pi \int \kappa_\nu S_\nu \,\mathrm{d}\nu$. Hence

$$\boxed{\int_0^\infty \kappa_\nu \left(J_\nu - S_\nu\right) \mathrm{d}\nu = 0}$$

- Absorption and emission balance *in total*, not at each frequency: a layer can absorb in the visible and emit in the infrared.
- The constraint involves J_ν, so it must be solved *together* with the transfer equation. The opacity's frequency dependence enters directly, and an inversion can arise naturally from the solution.
- The thermal (infrared) flux at the top is $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$: the interior heat plus the reradiated starlight. Interior heat comes from slow loss of formation energy (Jupiter emits about 1.7 times what it absorbs) or radioactivity (Earth); for Earth and close-in planets the stellar term dominates.

No general analytic solution exists; the **grey** atmosphere gives one.

Grey atmosphere heated from below

Assume LTE, no scattering, and frequency-independent opacity (grey). The energy comes from below only: from the interior of a giant planet ($T_{\rm int} = T_{\rm eff}$), or from a ground heated by sunlight that the atmosphere does not absorb ($T_{\rm int} = T_s$). Radiative equilibrium gives

$$J(\tau) = S(\tau) = B(T) = \frac{\sigma T^4}{\pi}.$$

This is exactly the grey atmosphere of Lecture 7: the moments of $\mu\,\mathrm{d}I/\mathrm{d}\tau = I - J$ give a constant flux $F = \sigma T_{\rm int}^4$ and $\mathrm{d}K/\mathrm{d}\tau = F/4\pi$; with $K = J/3$ and the surface boundary condition,

$$\boxed{T^4 = \frac{3}{4} T_{\rm int}^4 \left(\tau + \frac{2}{3}\right)}$$

- **Top of the atmosphere:** $T(0)^4 = T_{\rm int}^4/2$, the same $2^{-1/4}$ ratio as the simple greenhouse (Lecture 7).
- **Gradient:** differentiating with $\mathrm{d}\tau = -\kappa\,\mathrm{d}z$, $\mathrm{d}T/\mathrm{d}z = -\frac{3}{16}\kappa T_{\rm int}^4/T^3$ — the radiative-diffusion gradient. Large flux or large opacity gives a steep gradient.
- With the opacity's T, P dependence and hydrostatic equilibrium, $T(\tau)$ becomes a T–P profile.

Why T_{eq} is not enough: a myriad of effects

Without data, an exoplanet is labelled by its equilibrium temperature, computable without knowing anything about its atmosphere. The Solar System shows how misleading that can be: Earth's T_{eq} is below freezing, yet it has liquid water; Venus's T_{eq} is barely higher, yet its surface is about 700 K. Real atmospheres contain many opacity sources that warm or cool them (Heng, Ch. 4):

- **Greenhouse warming:** gases such as H_2O, CO_2 and CH_4 absorb the planet's *thermal* emission.
- **Anti-greenhouse cooling:** enhanced absorption, or scattering, of *starlight* high in the atmosphere.
- **Scattering greenhouse:** thermal emission is scattered back toward the planet.
- **Aerosols and clouds** combine all of these, and can cool or heat depending on their properties and where they sit.

The aim, following Pierrehumbert, is to understand this richness with simple, clear models that isolate one effect at a time — before, or instead of, a full simulation.

Why two bands? The dual-band approximation

Analytic profiles rest on the assumption that starlight and the planet's thermal emission occupy *distinctly different* wavelengths. Wien's law supports it:

$$\lambda_{\mathrm{peak}} \approx 0.5\,\mu\mathrm{m}\left(\frac{T}{6000\,\mathrm{K}}\right)^{-1} \qquad (\lambda_{\mathrm{peak}} T = 2898\,\mu\mathrm{m\,K}).$$

- Sun-like star: $\sim 0.5\,\mu$m. Earth (~ 300 K): $\sim 10\,\mu$m. Hot exoplanet (~ 1500 K): $\sim 2\,\mu$m.
- Heng calls the stellar range the **shortwave** (subscript S) and the planetary range the **longwave** (subscript L). They are usually the visible and the infrared — Seager's "vis" and "ir" bands below.
- **Caveat:** an M dwarf ($\lesssim 3000$ K) peaks in the near-infrared, so both bands can lie in the infrared; the separation is then less clean.

Within each band the opacity is replaced by a single mean value (a "double-gray" atmosphere), and each band can have its own scattering properties.

Heated from above and below: two bands

For an irradiated planet the starlight is absorbed *within* the atmosphere. The star is hotter than the planet, so starlight is absorbed at short wavelengths and re-emitted in the infrared. Treat two grey bands (Hansen 2008):

- **visible** ($I_{\rm vis}$, opacity $\kappa_{\rm vis}$): absorbed only, never emitted, $S_{\rm vis} = 0$;
- **infrared** ($I_{\rm ir}$, opacity $\kappa_{\rm ir}$): absorbed and re-emitted.

Radiative equilibrium now balances visible plus infrared absorption against infrared emission:

$$\kappa_{\rm vis} J_{\rm vis} + \kappa_{\rm ir} J_{\rm ir} = \kappa_{\rm ir} S_{\rm ir} \quad \Longrightarrow \quad \boxed{S_{\rm ir} = J_{\rm ir} + \gamma J_{\rm vis}, \qquad \gamma \equiv \frac{\kappa_{\rm vis}}{\kappa_{\rm ir}}}$$

With $\tau \equiv \tau_{\rm ir}$ and $\mathrm{d}\tau_{\rm vis} = \gamma \mathrm{d}\tau$, the stellar beam from direction μ_0 is only attenuated:

$$J_{\rm vis} = \frac{I_0}{4\pi} \mathrm{e}^{-\gamma\tau/\mu_0}, \qquad F_{\rm vis} = -\mu_0 I_0 \mathrm{e}^{-\gamma\tau/\mu_0} \quad \text{(downward)}.$$

Heated from above and below: the solution

Zeroth moment (IR): $\frac{1}{4\pi}\,\mathrm{d}F_{\mathrm{ir}}/\mathrm{d}\tau = J_{\mathrm{ir}} - S_{\mathrm{ir}} = -\gamma J_{\mathrm{vis}}$, so

$$F_{\mathrm{ir}}(\tau) = \mu_0 I_0\,\mathrm{e}^{-\gamma\tau/\mu_0} + F_{\mathrm{int}}, \qquad F_{\mathrm{ir}} + F_{\mathrm{vis}} = F_{\mathrm{int}} = \sigma T_{\mathrm{int}}^4 = \text{const.}$$

First moment with $K = J/3$: $\mathrm{d}J_{\mathrm{ir}}/\mathrm{d}\tau = (3/4\pi)F_{\mathrm{ir}}$,

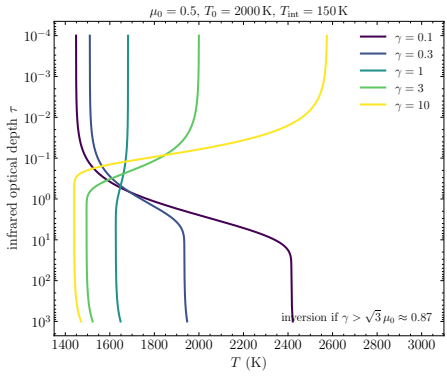
$$J_{\mathrm{ir}}(\tau) = \frac{3}{4\pi}\left[F_{\mathrm{int}}\,\tau - \frac{\mu_0^2}{\gamma} I_0\,\mathrm{e}^{-\gamma\tau/\mu_0}\right] + C.$$

Then $\sigma T^4/\pi = S_{\mathrm{ir}} = J_{\mathrm{ir}} + \gamma J_{\mathrm{vis}}$. Fixing C with the Eddington boundary condition $J_{\mathrm{ir}}(0) = F_{\mathrm{ir}}(0)/2\pi$ and writing $I_0 = \sigma T_0^4$,

$$\boxed{T^4 = \frac{3}{4} T_{\mathrm{int}}^4\left(\tau + \frac{2}{3}\right) + \frac{\mu_0 T_0^4}{4}\left[2 + \frac{3\mu_0}{\gamma} + \left(\frac{\gamma}{\mu_0} - \frac{3\mu_0}{\gamma}\right)\mathrm{e}^{-\gamma\tau/\mu_0}\right]}$$

Seager's Eq. (9.60) sets $\sigma T^4/\pi = J_{\mathrm{ir}}$, dropping the γJ_{vis} term required by her own Eq. (9.48). Her Eq. (9.62), after Hansen (2008), fixes C from the exact emergent flux rather than the Eddington boundary condition, which adds terms in $\ln(1+\gamma/\mu_0)$; the behaviour is the same.

What the irradiated profile shows



The boxed solution for several γ. Script

`scripts/make_irradiated_grey_profiles.py`.

- **Deep layers** ($\gamma\tau/\mu_0 \gg 1$): the starlight has been absorbed; $T^4 \to \frac{3}{4} T_{\mathrm{int}}^4 \tau + \mathrm{const}$ — the interior-heated profile, nearly isothermal when T_{int} is small.
- **Visible transparent** ($\gamma \to 0$): the profile tends to $\frac{3}{4}(T_{\mathrm{int}}^4 + \mu_0 T_0^4)(\tau + \frac{2}{3})$, as if all the flux came from below — the ground-heated case.
- **Thermal inversion** at the top when

$$\gamma > \sqrt{3}\,\mu_0,$$

  i.e. when starlight is absorbed higher up than the infrared can escape (strong optical absorbers: TiO/VO, metals, hazes).
- Smaller μ_0 (grazing starlight, near the terminator) deposits the energy higher and lowers the deep temperature.

Irradiated grey T–P profiles

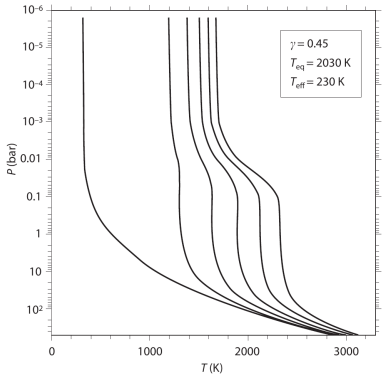


Figure 3: Grey-atmosphere T–P profiles for $\mu_0 = 0.0$–1.0 (left to right), $\gamma = 0.45$. Seager (2010), Fig. 9.7, after Hansen (2008).

- Converting τ to P with hydrostatic equilibrium, $\tau \simeq \kappa_m P/g$, turns $T(\tau)$ into a T–P profile.
- With $\gamma = 0.45 < \sqrt{3}\,\mu_0$ (for most μ_0) there is no inversion: an isothermal upper atmosphere, a transition where the starlight is absorbed ($\gamma\tau/\mu_0 \sim 1$), and a deep interior-heated part.
- A larger μ_0 (starlight closer to vertical) penetrates deeper and gives a hotter profile.
- Such analytic profiles are the usual **starting guess** for full radiative-convective models, and are still widely used for interpreting exoplanet spectra.

Scattering in a band: one parameter, and the Bond albedo

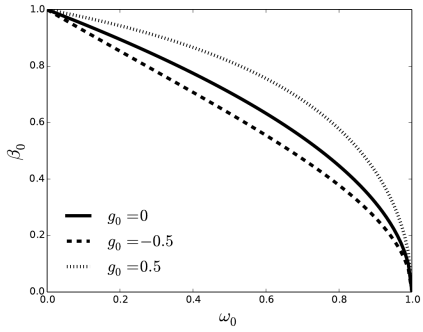


Figure 4: The scattering parameter versus ω_0 for isotropic ($g_0 = 0$), forward and backward scattering. Heng (2017), Fig. 4.1.

In the two-stream equations the single-scattering albedo ω_0 and the asymmetry factor g_0 never appear separately, only as the **scattering parameter**

$$\beta_0 \equiv \left(\frac{1-\omega_0}{1-\omega_0 g_0} \right)^{1/2},$$

with $\beta_0 = 1$ for pure absorption and 0 for pure scattering.

For starlight falling on an opaque atmosphere the two-stream solution gives the reflected fraction; averaged over the stellar spectrum,

$$\boxed{A_B = \frac{1-\beta_{S0}}{1+\beta_{S0}}}$$

- With $g_0 = 0$ this is the reflectivity ζ_-/ζ_+ of a thick slab in the two-stream test case (supplementary notes): 0.82, 0.52, 0.17 for $\omega_0 = 0.99, 0.9, 0.5$.
- Forward scattering ($g_0 > 0$) raises β_0: a cloud of forward-scattering particles is darker than its ω_0 alone suggests.

Where is the starlight absorbed? The photon deposition depth

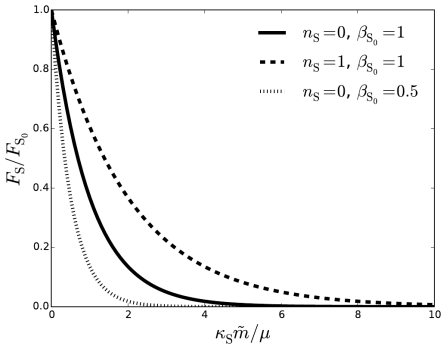


Figure 5: Generalized Beer's law. Scattering ($\beta_{S0} < 1$) makes the starlight fade faster; an opacity that falls with height ($n_S = 1$) lets it reach deeper. Heng (2017), Fig. 4.2.

Use the column mass $\tilde{m} = P/g$ as the depth coordinate. For a constant shortwave opacity κ_S (per unit mass) the incident stellar flux decays as a **generalized Beer's law**,

$$F_S = F_{S0} \exp\Big(- \frac{\kappa_S \tilde{m}}{|\mu| \beta_{S0}} \Big), \qquad F_{S0} = |\mu| \sigma T_{\rm irr}^4,$$

where $T_{\rm irr} = \sqrt{2} T_{\rm eq}$ is the substellar **irradiation temperature**. Scattering ($\beta_{S0} < 1$) shortens the path.

Averaged over the dayside, the flux falls as $2E_3(\kappa_S \tilde{m}/\beta_{S0})$, by $1/\mathrm{e}$ at $\kappa_S \tilde{m}/\beta_{S0} \approx 0.63$: the **photon deposition depth**

$$P_D \approx \frac{0.63\, g\, \beta_{S0}}{\kappa_S} = \frac{0.63\, g}{\kappa_S} \frac{1 - A_{\rm B}}{1 + A_{\rm B}}$$

Stronger scattering (higher albedo), or backward scattering, deposits the starlight *higher* (lower P).

An analytic model for controlled experiments

Heng derives the global-mean T–P profile of a dual-band atmosphere with scattering in both bands (Heng et al. 2014). Its inputs are few:

- $T_{\rm irr}$ (irradiation), $T_{\rm int}$ (interior heat), g;
- the shortwave opacity κ_S and the longwave opacity

$$\kappa_L = \kappa_0 + \kappa_{\rm CIA} \frac{\tilde{m}}{\tilde{m}_0},$$

  a constant part for greenhouse gases (H_2O, CH_4, CO) plus a part growing with pressure for collision-induced absorption;
- the scattering parameters β_{S0} and β_{L0}.

The solution is a sum of terms: an interior term, which is **Milne's solution** generalized to scattering, and stellar-heating terms containing E_2, E_3, E_4 of $\kappa_S \tilde{m}/\beta_{S0}$.

Fiducial model: $T_{\rm int} = 0$, $T_{\rm irr} = 1200\,{\rm K}$, $g = 10^3\,{\rm cm\,s^{-2}}$, $\kappa_S = 0.01$, $\kappa_0 = 0.02\,{\rm cm^2\,g^{-1}}$, no CIA, no scattering. Changing *one* parameter at a time isolates its effect, like a laboratory experiment.

The model sets the absorption, flux and Planck mean opacities equal; recovering Milne's solution, which needs the Rosseland mean, then implies these arithmetic means equal a harmonic one — the price of an analytic model.

Greenhouse warming and collision-induced absorption

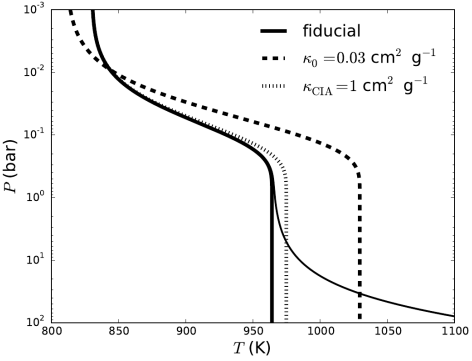


Figure 6: Fiducial model (thick solid), larger longwave opacity (dashed), CIA switched on (dotted); thin solid: fiducial with $T_{\rm int} = 150$ K. Heng (2017), Fig. 4.3.

- **Generic shape:** T rises with pressure, then turns *isothermal* below the depth where starlight is used up. With interior heat a deep "tail" appears: Milne's solution.
- Without an atmosphere the planet would be at $T_{\rm irr}/\sqrt{2} \approx 850$ K. The deep atmosphere is hotter: starlight enters easily, thermal emission escapes with difficulty — the **greenhouse effect**.
- **More longwave opacity** (dashed): stronger greenhouse; the lower atmosphere warms and the upper atmosphere cools.
- **Collision-induced absorption** (dotted): H_2 or N_2 have no permanent dipole, but in collisions at $\gtrsim 1$ bar a transient pair absorbs. A two-body process, so $\tau_{\rm CIA} \propto P^2$; it is a broad continuum and warms the deep layers most. First seen in Uranus and Neptune.

Anti-greenhouse cooling and thermal inversions

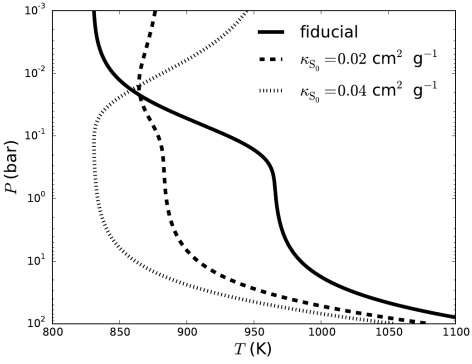


Figure 7: Increasing the shortwave opacity: $\kappa_S = \kappa_0$ (dashed) and $\kappa_S = 2\kappa_0$ (dotted), with $T_{\rm int} = 150$ K. Heng (2017), Fig. 4.4.

- Add absorbers of *starlight* until the shortwave opacity exceeds the longwave one: the **anti-greenhouse effect**. The upper atmosphere warms, the lower atmosphere cools.
- The isothermal deep part starts at lower pressure, because starlight is deposited higher up ($P_D \propto 1/\kappa_S$).
- For large enough κ_S/κ_L the profile develops a **thermal inversion** — the same physics as the criterion $\gamma > \sqrt{3}\,\mu_0$ found earlier.
- Earth: O_3 absorbing solar UV. Hot Jupiters: TiO/VO, by analogy with brown dwarfs, and atomic metals; the absorber is still debated.
- A warm upper atmosphere also shortens chemical timescales, which can undo disequilibrium chemistry — inversions matter for interpreting abundances.

Albedo and the scattering greenhouse

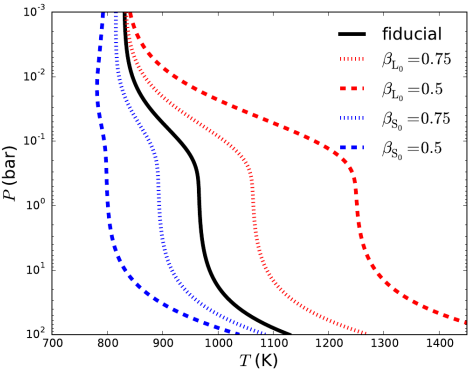


Figure 8: Shortwave scattering (blue, $\beta_{S0} = 0.75, 0.5$) and longwave scattering (red, $\beta_{L0} = 0.75, 0.5$), with $T_{\rm int} = 150$ K. Heng (2017), Fig. 4.5.

- **Shortwave scattering** raises the Bond albedo: $\beta_{S0} = 0.75$ and 0.5 give $A_{\rm B} = 1/7$ and $1/3$. Like the anti-greenhouse effect it warms the top relative to the bottom, but in addition the *whole* profile shifts cooler: the absorbed energy is reduced by $(1 - A_{\rm B})$.
- **Longwave scattering** does the opposite: thermal emission trying to escape is partly scattered back, and the atmosphere is warmer everywhere — the **scattering greenhouse**. Scattering in the infrared needs particles of a micron or larger.
- **Clouds and aerosols** can do all of these at once — greenhouse and anti-greenhouse, by absorption and by scattering. The effects may reinforce or cancel, which is why clouds complicate every interpretation.

Summary of the effects

effect	cause	effect on the T–P profile
greenhouse warming	more longwave opacity	lower layers warmer, upper cooler
collision-induced absorption	$\tau_{\mathrm{CIA}} \propto P^2$ (H_2, N_2)	deep layers warmer
anti-greenhouse cooling	more shortwave opacity	upper layers warmer, lower cooler; inversion if strong enough
higher Bond albedo	shortwave scattering	anti-greenhouse + whole profile cooler
scattering greenhouse	longwave scattering	warmer everywhere
interior heat	$T_{\mathrm{int}} > 0$	deep Milne “tail”

Each of these is a single parameter in the analytic model, which is why such models remain the standard tool for building intuition before a full radiative–convective calculation.

Convection basics

Convection transports energy by **bulk vertical motion**: heat flows from hot to cold regions with the moving gas. It is driven by the temperature gradient and enabled by gravity (buoyancy).

- A parcel heated from below becomes lighter than its surroundings and rises. At lower ambient pressure it expands and releases heat; now cooler, it sinks, is reheated, and the cycle repeats.
- Everyday examples: boiling water, lava lamps, the shimmering air above hot pavement.
- The Sun's **granulation**: bright centres are rising columns of hot gas, dark lanes are cooled gas sinking; cells are $\sim 1000\,\mathrm{km}$ across.

The dry adiabatic lapse rate

First law for a parcel, $\mathrm{d}U = \mathrm{d}Q - \mathrm{d}W$, with $\mathrm{d}Q = 0$ (adiabatic), $\mathrm{d}U = mc_v\,\mathrm{d}T = m(c_p - R_s)\,\mathrm{d}T$ and $\mathrm{d}W = P\,\mathrm{d}V$. The ideal gas law $PV = mR_sT$ gives $P\,\mathrm{d}V = mR_s\,\mathrm{d}T - (m/\rho)\,\mathrm{d}P$, so

$$\frac{\mathrm{d}T}{\mathrm{d}P} = \frac{1}{\rho c_p} \xrightarrow{\mathrm{d}P = -\rho g\,\mathrm{d}z} \boxed{\Gamma \equiv -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}}$$

- Γ is the **adiabatic lapse rate** (positive when temperature falls with height). A rising parcel shares the pressure of its surroundings but follows its own T and ρ.
- In optical depth, $\mathrm{d}T/\mathrm{d}\tau = g/(\kappa_m c_p)$.
- In pressure: $\mathrm{d}T/T = (R_s/c_p)\,\mathrm{d}P/P$, so along an adiabat

$$T \propto P^{R_s/c_p}, \qquad R_s/c_p = 2/7 \text{ for a diatomic ideal gas.}$$

On a log–log T–P diagram an adiabat is a straight line.

The moist adiabat

When a condensable vapour condenses in the rising parcel, its latent heat warms the parcel: $\mathrm{d}Q = -L\,\mathrm{d}m_s$, with m_s the vapour mass per unit mass of non-condensable gas and L the specific latent heat. Keeping $\mathrm{d}Q$ in the first law,

$$\boxed{\Gamma_{\mathrm{moist}} = \frac{g}{c_p}\,\frac{1}{1+(L/c_p)(\mathrm{d}m_s/\mathrm{d}T)}} \qquad (\mathrm{d}m_s/\mathrm{d}T > 0,\ \text{so}\ \Gamma_{\mathrm{moist}} < \Gamma).$$

- Earth's troposphere: dry $\Gamma \simeq 9.8\,\mathrm{K\,km^{-1}}$, moist $\simeq 5\,\mathrm{K\,km^{-1}}$; an average $6.5\,\mathrm{K\,km^{-1}}$ is typical since convection is partly dry, partly moist.
- The observed profile is *stable* against dry convection but *unstable* against moist convection: latent heat is what drives Earth's convection.
- Exoplanets with other temperatures condense other species (e.g. NH_3, CH_4, silicates, iron); the same formula applies with their L and saturation curves.

Conceptual overview

A radiative–convective model solves **three equations for three unknowns** — the temperature T, the pressure P, and the radiation field I_ν:

1. hydrostatic equilibrium, $\mathrm{d}P/\mathrm{d}\tau = g/\kappa_m$;
2. the transfer equation, $\mu\,\mathrm{d}I_\nu/\mathrm{d}\tau = I_\nu - S_\nu$;
3. radiative (and convective) equilibrium, $\int \kappa_\nu (J_\nu - S_\nu)\,\mathrm{d}\nu = 0$.

Procedure.

- Start from a guess, e.g. the grey profile with T- and P-dependent opacities.
- Solve the three equations numerically, applying temperature corrections until all are satisfied.
- Check every layer for convective stability. Where it is unstable, replace the profile by the adiabat and iterate again, so that the radiative part joins the adiabat smoothly.

A radiative-equilibrium atmosphere: inputs and subtleties

Outputs: $I(\tau, \mu, \nu)$, $T(\tau)$, $P(\tau)$ — the T–P profile and the emergent spectrum together.

Inputs: the interior temperature T_{int} (from evolution models, or free), the surface gravity g (known for transiting planets), the stellar irradiation, and the opacities — which need the composition, from chemical equilibrium or a photochemical model.

Subtleties:

- The equations are tightly coupled. A new $T(\tau)$ changes the chemistry, hence the number densities, the pressure, κ_ν and ε_ν, hence the radiation field everywhere.
- Scattering makes the transfer equation non-local: stellar photons can travel deep before they heat the gas, coupling the top of the atmosphere to the deep layers.
- Hence iteration from a good starting profile, typically a grey solution.

Radiative–convective equilibrium

Take the radiative-equilibrium profile and test each layer with the Schwarzschild criterion, $|\mathrm{d}T/\mathrm{d}z|_{\mathrm{rad}} > |\mathrm{d}T/\mathrm{d}z|_{\mathrm{ad}}$.

- Usually convection is assumed so efficient that, where it occurs, the gradient *is* the adiabat. The adiabat fixes only the slope; its level is set by matching the radiative temperature at the top of the convection zone.
- Where energy is shared, the net flux is split: $F_{\mathrm{rad}} + F_{\mathrm{conv}} = \sigma T_{\mathrm{int}}^4$ (deep layers, where the starlight has been absorbed).
- The convective flux is

$$F_{\mathrm{conv}} = \bar{v}\,\rho c_p\,\Delta T, \qquad \Delta T = \left[\left(-\frac{\mathrm{d}T}{\mathrm{d}z}\right) - \left(-\frac{\mathrm{d}T}{\mathrm{d}z}\right)_{\mathrm{c}}\right]\Delta z,$$

  the excess heat of a parcel that travels Δz at mean speed $\bar{v}$ before merging with its surroundings.
- $\bar{v}$ and Δz come from **mixing-length theory**, a local 1D approximation from stellar physics: the mixing length $l \sim H$ is the distance a bubble rises before releasing its heat.

Seager's Eqs. (9.75)–(9.76) write the constant flux as $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$ and $\sigma T_{\mathrm{eff}}^4 + \mu_0 I_0$. The net (up minus down) flux that is constant with depth is $\sigma T_{\mathrm{int}}^4$; $\sigma T_{\mathrm{int}}^4 + \mu_0 I_0$ is the outgoing thermal flux at the top.

Is the deep atmosphere convective? Milne's solution and CIA

Write the Schwarzschild criterion in logarithmic form: the atmosphere is stable if

$$\frac{\partial \ln T}{\partial \ln P} \leq \nabla_{\rm ad} = \frac{R_s}{c_p} = \frac{2}{2+n_{\rm dof}},$$

with $n_{\rm dof}$ the degrees of freedom of the dominant gas: for a diatomic gas, $\nabla_{\rm ad} = 2/7$ (translation and rotation) or $1/4$ (with vibration). H_2 (giant planets) and N_2 (Earth, Titan) are diatomic.

Deep down, interior heat gives Milne's solution, $T^4 \propto \tau$:

- **Constant opacity:** $\tau \propto P$, so $T \propto P^{1/4}$ and $\partial \ln T/\partial \ln P \to 1/4 \leq 2/7$: **stable**, radiative.
- **CIA dominates:** $\kappa \propto P$, so $\tau \propto P^2$, $T \propto P^{1/2}$ and $\partial \ln T/\partial \ln P \to 1/2 > 2/7$: **unconditionally convective**.

The result does not depend on the strength of absorption, on scattering, or on gravity: when collision-induced absorption operates deep down, the interior of a gas giant must convect. Quantum-mechanical opacities, via Schwarzschild's criterion, decide a macroscopic flow.

Higher up, irradiation usually stabilizes the profile; but with a strong greenhouse (shortwave opacity much smaller than longwave), *detached* convective layers can appear between stable zones (Heng, problem 4.9.2).

Forward and inverse problems

- The **forward problem**: physics → T–P profile → spectrum, by iterating the coupled equations.
- The **inverse problem**: spectrum → T–P profile. For Solar System planets, high-quality spectra allow a fiducial profile to be inferred and then perturbed (in T and abundances) until the data are fit.
- Exoplanet spectra are not good enough to infer a unique fiducial profile. Instead, map out the range of T–P profiles and abundances the data allow.
- Full forward models are too slow to run millions of times, so Madhusudhan & Seager (2009) used a **parametric T–P profile**, running tens of thousands of models that satisfy hydrostatic equilibrium and global energy balance. The result constrains the T–P range, molecular abundances, albedo, day–night redistribution and effective temperature.

A parametric T–P profile

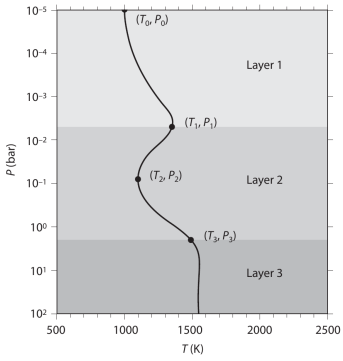


Figure 9: The parametric P–T profile, with a possible inversion in layer 2. Seager (2010), Fig. 9.8, after Madhusudhan & Seager (2009).

$$P = P_0\, e^{\alpha (T - T_0)^{\beta}} \quad \text{in layers 1 and 2,} \qquad T = T_3 \text{ in layer 3,}$$

with $\beta = 1/2$ and continuity at the layer boundaries; with $P_0 = 10^{-5}$ bar this leaves **six free parameters**.

- **Above P_0:** transparent at all wavelengths; no spectral features.
- **Layer 1:** no inversion; heated from below, cooling upward.
- **Layer 2:** where most features form; radiative (and perhaps dynamical) processes; an **inversion** may form with strong irradiation and absorbers.
- **Layer 3:** high optical depth, radiative diffusion, nearly isothermal — heated by the interior, not the star. Below it, convection.
- Cooler planets: no layer 3, layer 2 extends deeper, and the radiative–convective boundary is higher (an adiabat in layers 2–3).

Take-home messages

Leaky and multilayer greenhouse models. $T_s = [2/(2-\alpha)]^{1/4} T_e$ for one layer with infrared absorptivity α; $T_s = (n+1)^{1/4} T_e$ for n fully opaque layers.

Convection versus radiation. Convection occurs when $|\mathrm{d}T/\mathrm{d}z|_{\mathrm{atm}} > g/c_p$. High opacity, large flux, or large c_p favour instability; temperature inversions are stable.

Radiative equilibrium. $\int \kappa_\nu (J_\nu - S_\nu)\mathrm{d}\nu = 0$. Grey, heated from below: $T^4 = \frac{3}{4} T_{\mathrm{int}}^4 (\tau + \frac{2}{3})$. Heated from above too: two bands with $\gamma = \kappa_{\mathrm{vis}}/\kappa_{\mathrm{ir}}$; an inversion forms when $\gamma > \sqrt{3}\,\mu_0$.

Greenhouse and anti-greenhouse. $A_{\mathrm{B}} = (1-\beta_{S0})/(1+\beta_{S0})$; starlight is deposited near $P_D \approx 0.63\, g\beta_{S0}/\kappa_S$. Longwave opacity or CIA warms the deep layers; shortwave opacity warms the top and can invert the profile; albedo cools; infrared scattering warms. With CIA the deep Milne profile ($T \propto P^{1/2}$) must convect.

Adiabat and models. $\Gamma = g/c_p$ (9.8 K/km dry on Earth), smaller when vapour condenses; $T \propto P^{R_s/c_p}$. Full models solve hydrostatic, transfer and radiative–convective equilibrium together by iteration; retrievals use parametric T–P profiles.

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